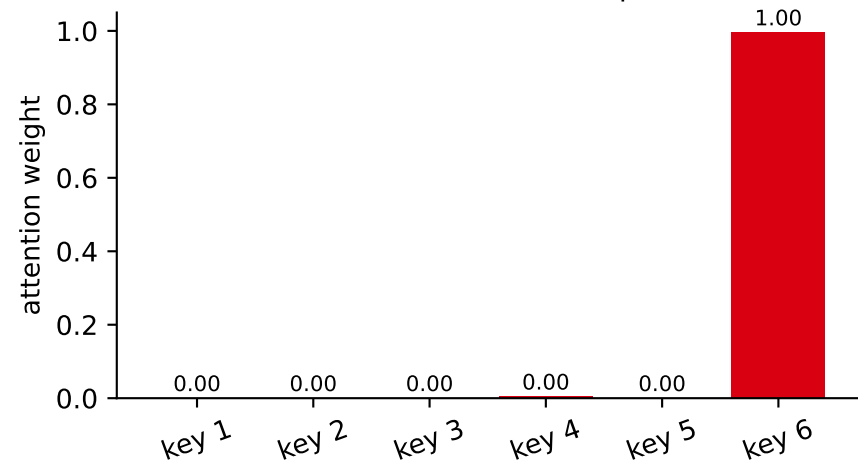


Why  $\sqrt{d_k}$ : big dimensions make dot products large  $\rightarrow$  softmax collapses to a near-hard argmax ( $d_k = 64$ )

softmax of raw  $QK^\top$  (scores spread  $\approx \pm 5$ )



softmax of  $QK^\top / \sqrt{d_k}$  ( $\approx \pm 1$ )

